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Extreme Value Distribution of

Global Temperature

by

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abstract

Global climate extremes are becoming increasingly common, and an accurate estimate of frequency and distribution of these events can significantly aid in developmental policy-making and all sorts of human activities.

The objective of this study is to fit the GEV model for global temperature data from NOAA, and try to obtain an efficient probabilistic approach from the family of extreme value distribution models, which can suitably model the uncertainties pattern in extreme global temperature event.

Keywords: Extreme Value Distribution, Global Temperature, GEV, EVT, Extreme Value Theory

Contents

1	Introduction	1
2	Extreme Value Model	4
3	Generalized Extreme Value Distribution	9
4	Maximum Likelihood Estimate	11
5	Data	13
6	Result	15
7	Conclusions	23
8	Recommendation	25
9	Appendix	28

Chapter 1

Introduction

When the Swedish girl, Greta Thunberg, grasped the global attention for climate change, and more and more students joined her for a "Fridays for Future", statisticians have already focused on extreme weather and tried to use Generalized Extreme value Distribution (GEV) and Geo-statistics concepts to fit the model and try to come up with useful suggestions for global leaders.

However, whether a particular extreme event is due to so called Climate Change is hard to determine because of the natural variability. [[4]]. Just as [7] stated, the evidence of potential changed in extreme events is mixed, mainly due to a lack of data. The most robust conclusions with regards to the change in extreme weather and climate events are according to the IPCC report: (a) an

increased probability of extreme warm days and decreased probability of extreme cold days and (b) an increased chance of drought for mid-continental areas during summer with increasing CO_2 .

Although statistical modeling for extreme climate events is not new, there are several difficulties as below,[7]: The evaluation of many types of extreme events is made difficult because of the small spatial and temporal scales that damaging extreme events often have; therefore, intense events are not well-represented (or not represented at all) in model-simulated climates, in particular for the coupled climate models with a typical horizontal resolution of 300 to 500 km. In addition, there is often a mismatch between the scales resolved in these models and those of the observed data.

Re-analysis has helped in this regard, but re-analysis is not pre se the sole answer. [7], because the model used for the re-analysis rely on sub-grid parametrizations almost as heavily as climate models do. That is to say, statistical modeling of extremes may be based on limited families of distributions for maximum that satisfies the requirement of max-stability.

As a result, this project aims at presenting the scale of the global temperature from 2010 to 2019, with science mapping of Python packages: Matplotlib, Basemap, NetCDF and sciPy, and then fit the long term mean of observation

data of 2019 global temperature provided by NOAA(National Oceanic and Atmospheric Administration) by estimating the location, scale and shape parameters.

Lastly, from the basic time-series plots, this project presents the global temperature patterns from 2010 to 2019, except for 2012 and 2016 because the days of the year in the data set is not consistent. It is obvious that the pattern has increasingly large fluctuations since 2015 onward. The long term means of daily maximum temperature in 2019 follows an extreme value distribution, with highly left-tail skewed, although the GEV seems not fit very well.

Chapter 2

Extreme Value Model

Basically the Fisher-Tippett Theorem states that for an i, i, d sequence of random variables, for example a time-series segregated into blocks, we can construct another sequence called M consisting of the maxima of each random variable (or minima of each), and if the (normalized) distribution of M converges asymptotically under the condition that sample size n tends to infinity, it must converge to one of three distributions called Gumbel, Fréchet and (reverse) Weibull.

Below is the PDF of the GEV distribution, citing from Wikipedia:

This project would talk about two approaches to generate GEV models: "Block Maxima/Annual Maximum Series(AMS)" and "Threshold Exceedances/Partial Duration Series (PDS)".

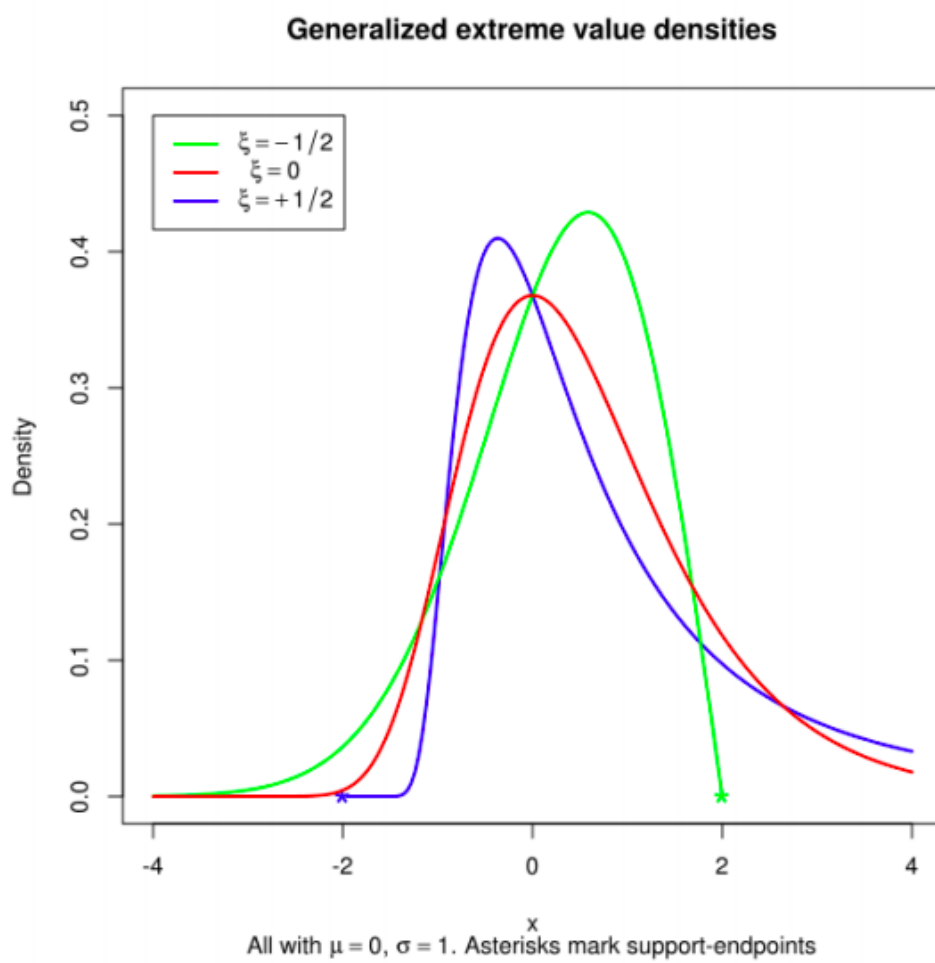


Figure 2.1: Generalized Extreme Value Distribution. Source: Wikipedia

What is Block Maxima? The block maxima method directly extends the Fisher-Tippett Theorem given above and the assumption is that each block forms a random i, i, d sample from which an extreme value distribution can be fitted. So in other words, we will fit a GEV distribution to the maxima sampled from each block. In this project, each block is one year.

Block Maxima can be used to answer such question as: In a given block size, for example, a year, how likely is an exceedance of a specified threshold? What threshold can be expected to be exceeded, on average, every N blocks?

And what is PDS? We are often more interested in the PDS-based Extreme Value Theory(hereafter as EVT) statistics when we are more concerned about what is the threshold value or exceedances would become harmful, instead of simply understanding the maxima in a specific block. PDS is also termed as “Peaks-over-Threshold” (PoT) method.

Below is the definition and PDF for PDS, cited from Wikipedia:

The difficulties for applying PDS are mainly the following two: One is hard to choose the appropriate threshold. Usually it is preferable to choose the minimum of AMS as the threshold. The second challenge is the correlation of the data. It is not easy to make sure the samples are independent to each other.

Block Maxima has the advantages such as easy to apply and interpret, while

The standard cumulative distribution function (cdf) of the GPD is defined by^[5]

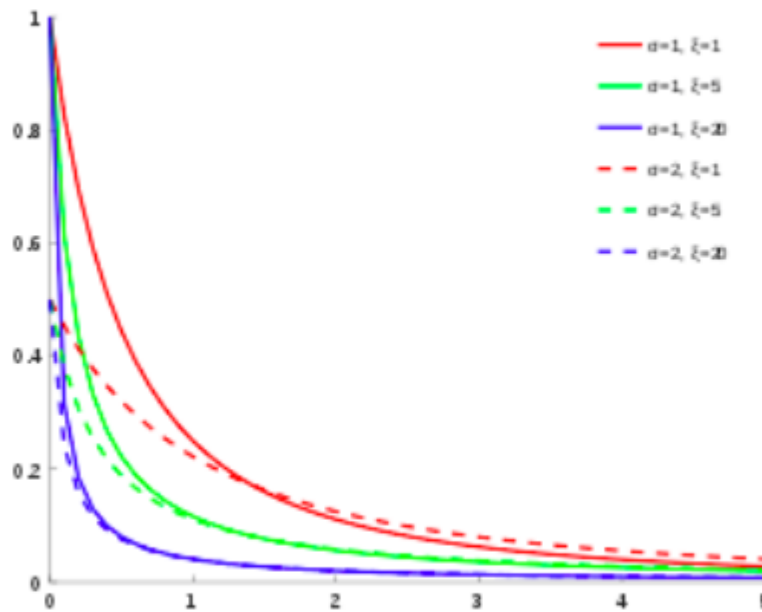
$$F_{\xi}(z) = \begin{cases} 1 - (1 + \xi z)^{-1/\xi} & \text{for } \xi \neq 0, \\ 1 - e^{-z} & \text{for } \xi = 0. \end{cases}$$

where the support is $z \geq 0$ for $\xi \geq 0$ and $0 \leq z \leq -1/\xi$ for $\xi < 0$. The corresponding probability density function (pdf) is

$$f_{\xi}(z) = \begin{cases} \frac{1}{\sigma} (\xi z + 1)^{-\frac{\xi+1}{\xi}} & \text{for } \xi \neq 0, \\ \frac{1}{\sigma} e^{-z} & \text{for } \xi = 0. \end{cases}$$

Generalized Pareto distribution

Probability density function



GPD distribution functions for $\mu = 0$ and different values of σ and ξ

Figure 2.2: Generalized Extreme Value Distribution. Source: Wikipedia

PDS uses data more efficiently. In reality, these two are pretty similar in their functionality. In [\[4\]](#), it states that it was proved that these two approaches converge at high return periods or rare events.

The assumptions for EVT is as below:

Firstly, assume i, i, d random variables: Each sample is independent from all other samples. Secondly, identically distributed: Each sample is taken from the same underlying distribution, and we assume no seasonality in the time-series. Also, we assume stationarity in the time-series. Lastly, randomness. Values sampled from a probability distribution, and varies due to chance. Measurements are unbiased estimator of the desired quantity.

Chapter 3

Generalized Extreme Value Distribution

From Fisher-Tippett Theorem, the "Central Limit Theorem" for Generalized Extreme Value distribution (hereafter as GEV) has the mathematical formula as:

Consider the set of i, i, d random variables: $X_1, X_2, \dots, X_n$, each X_i is sampled from an unknown underlying distribution function F .

Let M_n equals to $\text{Max}(X_1, X_2, \dots, X_n)$, given certain limiting conditions met, there exists a set of real numbers (a_n, b_n) , $a_n \geq 0$ such that $(M_n - b_n)/a_n$ has a non-degenerate distribution, which leads to $M_n \text{ GEV}(\mu, \sigma, \xi)$.

To start an extreme value analysis, we may use Python's "sciPy" package to

generate a GEV for the global temperature data we have.

From the Extreme Value Theorem, the GEV is the limit-distribution of the block-maxima of any randomly distributed variable. There are three parameters in the GEV distribution: location, scale, and shape parameters, and the Block length for the data here is 365 days.

We can extract the maximum temperature value from each year, totaling 30 years (1981-2010), and we obtained the Annual Maxima Series (AMS) as in Table 1.

Chapter 4

Maximum Likelihood Estimate

After obtaining the AMS, we are ready to fit a GEV to it.

Here we used L-moments to fit the GEV. L-moments supposes that we have a series of ranked statistics: $X_1, X_2, \dots, X_n$, the n^{th} L-moment may be defined as:

$\lambda_n(X) = (1/n) \sum_{i=0}^{(n-1)} (-1)^{i(n-1)} C_i E[X_{n-i}]$, where E denotes expectation.

For a sample of size s , the sample L-moment may be computed as:

$$\hat{\lambda}_n(X_n) = (1/(n^s C_n)) \sum_{i=1}^s (\sum_{b=0}^n (-1)^b {}^n C_b {}^{i-1} C_{n-b} {}^{s-1} C_b) X_i.$$

Probability distribution parameters may be described as a function of the L-moments of the underlying distribution. [[4]]

Here we used Python's sciPy package and "gev.fit" module to obtain the three parameters of GEV distribution. The shape, location and scale parameters are

0.941830383418343, -2.509716860216998, and 5.291777014216436. And then we can fit the extreme value distribution PDF and compare the PDF with the histogram. The result is in FIGURE 5.

We can see the probability distribution of the 2018 CPC Global Daily Temperature is a extremely left-skewed data, and from the scatter plot of its PDF, we can also see some outliers. Hence, it is pretty clear that the CPC Global Temperature data from 1981 to 2010 follows a GEV distribution.

Chapter 5

Data

The data set used here is called "CPC Global Daily Temperature" from National Oceanic and Atmospheric Administration (NOAA). The NOAA Global Surface Temperature Dataset (NOAAGlobalTemp) is freely available from the NOAA website.[1] The temporal coverage for the data sets we use in this project include Daily Maximum Temperature data from 1979/01/01 to present and Long Term Means of daily, monthly for years 1981 to 2010.

The data description is in FIGURE 5.1 & FIGURE 5.2:


```

NetCDF Global Attributes:
  Conventions: 'CF-1.0'
  Source: 'ftp://ftp.cpc.ncep.noaa.gov/precip/wd52ws/global_temp/'
  References: 'https://www.esrl.noaa.gov/psd/data/gridded/data.cpc.globaltemp.html'
  version: 'V1.0'
  title: 'CPC GLOBAL TEMP V1.0'
  dataset_title: 'CPC GLOBAL TEMP'
  history: 'Updated 2019-09-10 20:53:55'
NetCDF dimension information:
  Name: lat
    size: 360
    type: dtype('float32')
    actual_range: array([ 89.75, -89.75], dtype=float32)
    long_name: 'Latitude'
    units: 'degrees_north'
    axis: 'Y'
    standard_name: 'latitude'
    coordinate_defines: 'center'
  Name: lon
    size: 720
    type: dtype('float32')
    long_name: 'Longitude'
    units: 'degrees_east'
    axis: 'X'
    standard_name: 'longitude'
    actual_range: array([2.5000e-01, 3.5975e+02], dtype=float32)
    coordinate_defines: 'center'
  Name: time
    size: 365
    type: dtype('float64')
    long_name: 'Time'
    units: 'hours since 1900-01-01 00:00:00'
    delta_t: '0000-00-01 00:00:00'
    avg_period: '0000-00-01 00:00:00'
    axis: 'T'
    standard_name: 'time'
    coordinate_defines: 'start'
    actual_range: array([1034376., 1043112.])

```

Figure 5.1: CPC Global Daily Temperature Data Description

```

NetCDF variable information:
  Name: tmax
    dimensions: ('time', 'lat', 'lon')
    size: 94608000
    type: dtype('float32')
    missing_value: -9.96921e+36
    units: 'degC'
    var_desc: 'Maximum Temperature'
    level_desc: 'Surface'
    statistic: 'Maximum'
    parent_stat: 'Other'
    long_name: 'Daily Maximum Temperature'
    cell_methods: 'time: mean'
    valid_range: array([-90., 50.], dtype=float32)
    avg_period: '0000-00-01 00:00:00'
    dataset: 'CPC Global Temperature'
    comments: 'GTS data and is gridded using the Shepard Algorithm'
    max_period: '6z to 6z'
    actual_range: array([-58.620934, 51.97627 ], dtype=float32)

```

Figure 5.2: CPC Global Daily Temperature Data Description

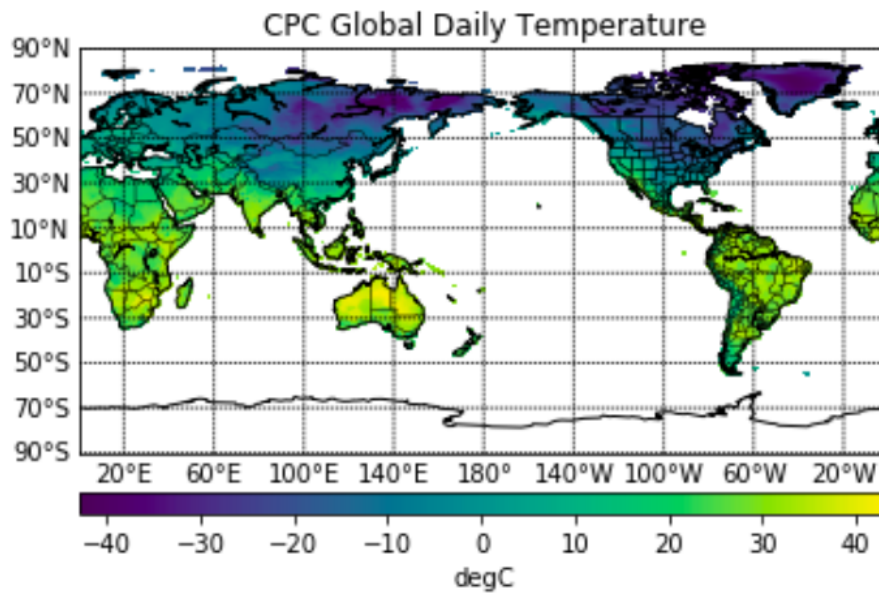
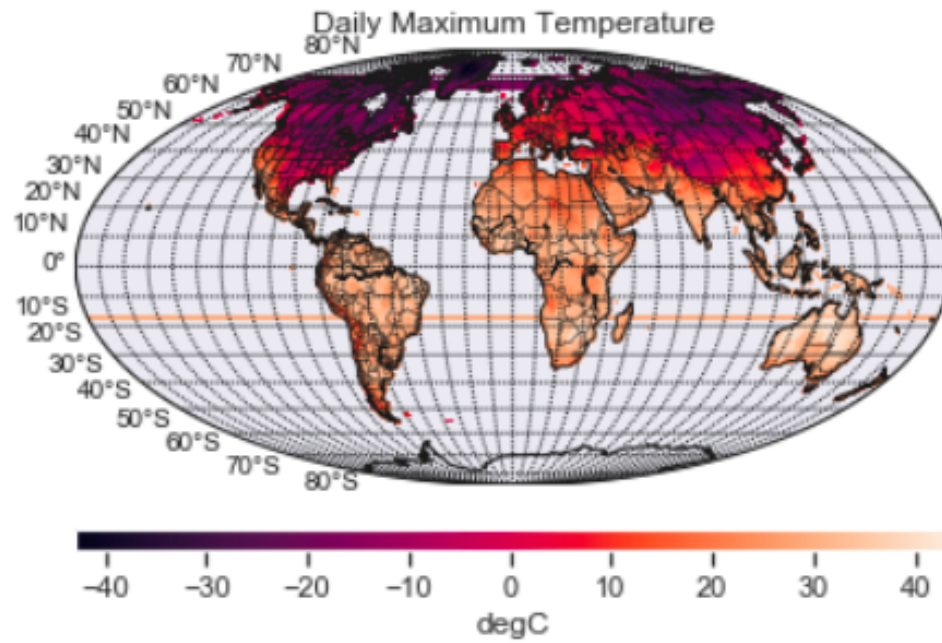
Chapter 6

Result

Table 1 is the Annual Maximum Series we generated from the data. The block is one year, which is 365 days, and the days per year is 365, and the number of year is 30, which is 1981 to 2010. So we have 30 AMS.

```
[ -19.11105919 -14.26672077 -10.2001276  -8.0826149  -5.97519922
  -2.66958022  -2.41479635  -2.58135891  -1.26270354  -1.08711851
  -0.89836979  -0.23375072   0.59504539   1.19994652  -0.08037828
  -1.43685365  -0.25511813   1.27147734   1.47244573  -9.98015308
   0.70499694   1.29618537  -7.11014462   1.51492488   2.47590113
  -1.28199124   1.91829848   1.92789495   0.03419649   3.09616828]
```

From the two plots below, we can see the scale of global daily temperature from 1981 to 2010. The data set used is in **Appendix [1]**.



And from the following Figures, we can see how the data fit the GEV distri-

bution. Figure 6.1, 6.2 shows it is a highly left-skewed extreme value distribution, and the scatter plot of the fitted GEV PDF shows that there are some outliers on the lower right side of the plot. From these plots, we can see clearly the global temperature rapidly increased within this 30 years.

Lastly, we can see the time-series plots over the years. We used the individual observation data sets of long term means of daily temperature from 2010 to 2019 in **Appendix [1]**. Through the trends, although for 2012 and 2016, due to the data sets have 366 days and for the consistency of comparison, we excluded these two years in this plot; for 2019, it has only 273 days, we can still see from the trend lines below that for 2017 and 2018, the fluctuation of global temperature is more radical than the years before. During winter times, which is in the bottom right of the plot, in 2013 and 2015, the temperature seems higher than other years at the same time.

We can see separate time-series plots for 2011,2013,2014,2015,2017, and 2018 in the following **Figure 6.4** and **6.5**.

The reason why 2012 and 2016 has not been included here is the days of the year for these two years is 366. And for 2019, it has only been 273 days in the data set available, but we can tell from **Figure 6.6** below that the global temperature in 2019 is obviously and rapidly increasing.

Shape Parameter is 0.941830383418343
Location Parameter is -2.509716860216998
Scale Parameter is 5.291777014216436

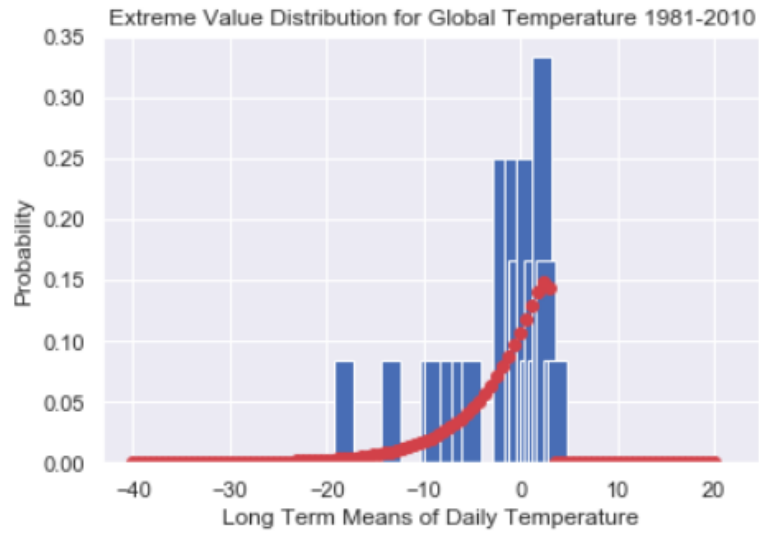


Figure 6.1: GEV for Long Term Means of Daily Global Temperature.

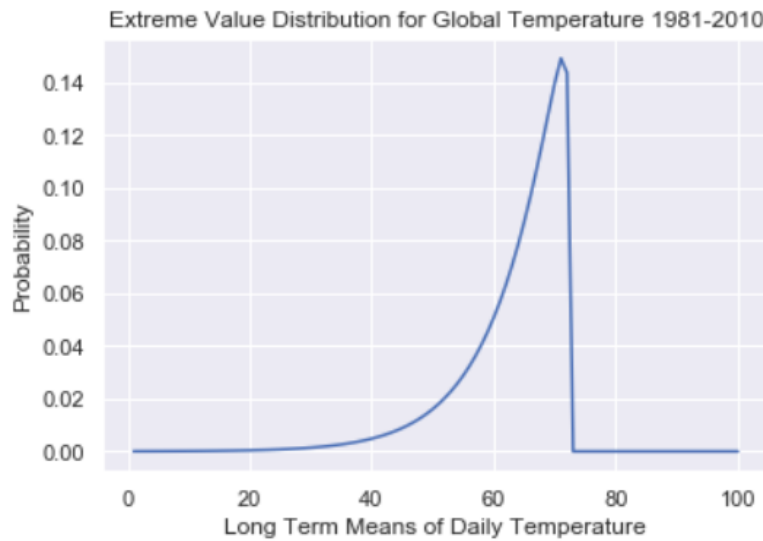


Figure 6.2: GEV for Long Term Means of Daily Global Temperature.

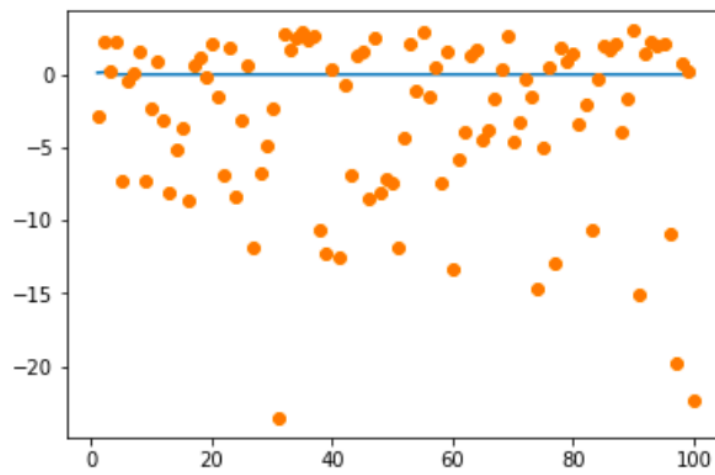


Figure 6.3: Scatter Plot of 2019 Global Temperature GEV.

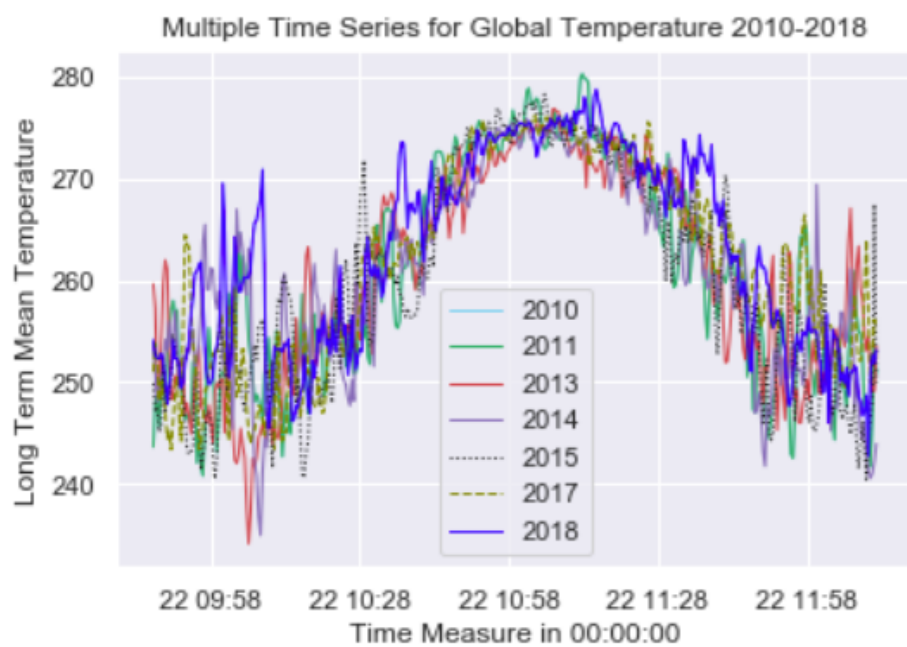


Figure 6.4: Time-series from 2010 to 2018.

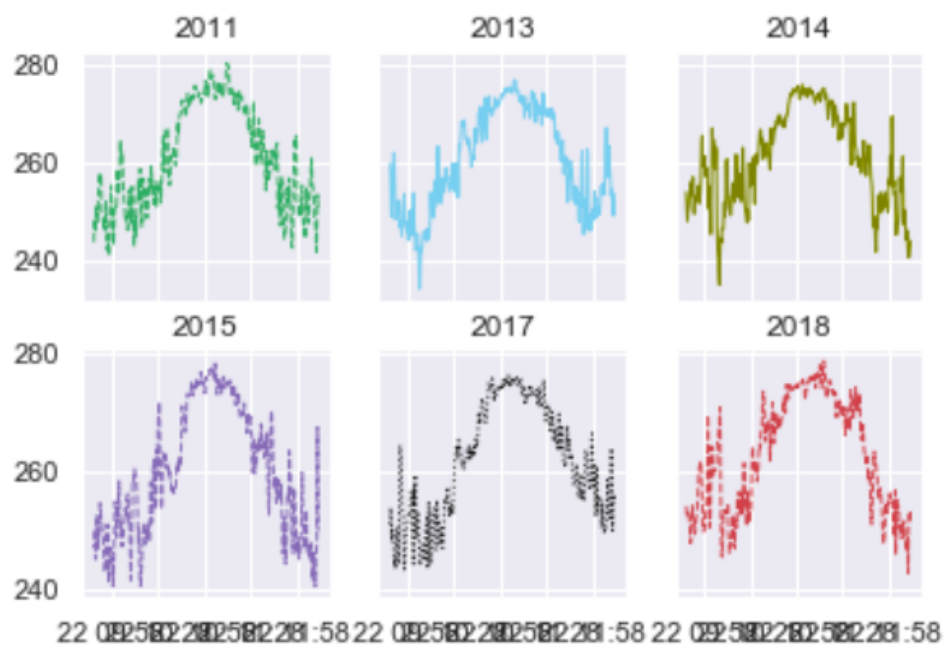


Figure 6.5: Separate time-series plots.

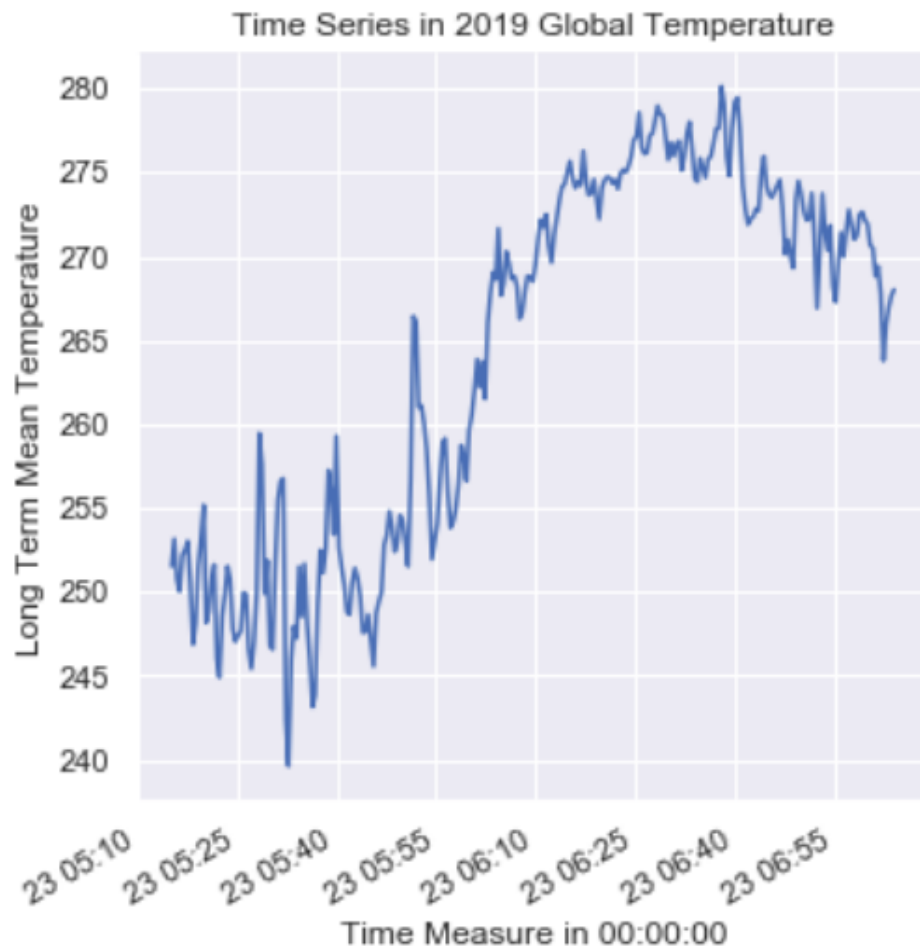


Figure 6.6: Global Temperature in 2019.

Chapter 7

Conclusions

In conclusion, the CPC Global Temperature data [**Appendix 1**] in this project, from the time-series plots in the figures above, because it is a whole year trend, it might not be crystal clear that the fluctuation in global temperature from year to year has been becoming more drastic, as the unit scale is not very visually observable. However, it is still presenting the trend that from 2015 to 2017, the maximum temperature for the summer and the minimum temperature for the winter are reaching to even higher and even lower level than the same times in previous years.

In addition, this project also used maximum likelihood method to estimate the three parameters of the GEV for the global temperature from 1981 to

2010.(**Figure 6.1 & 6.2**).

All science mapping in this project used Python packages: sciPy, Matplotlib, Numpy, Catopy, Basemap, NetCDF. And the IDE was Jupyter notebook.

We used three different data sets in the mapping:

One is the CPC Long Term Means of daily, monthly for years 1981 to 2010; The other is CPC Long Term Means of daily for 2019; and still another is CPC Individual Observation for the years 2010 to 2019.

From **Figure 6.2**, it is clearly shown that the data follows a strongly left-skewed Generalized Extreme Value Distribution.

Chapter 8

Recommendation

Extreme Value Theory(EVT) emerges very early, but its application on climatology seems have not yet been very popularized and easily accessible. Science mapping and Geo-statistics are also highly potential research areas that combine data analysis, data visualization, and highly cooperative with other disciplines in real life problems. There are a lot of variations in model fitting and prediction for extreme climate phenomena, which makes it harder to increase accuracy of the prediction and to suggest highly accountable solutions.

As a result, the more extensive, and deeper research endeavors in the relevant topics, and more cooperative discussions cross-disciplines would be great to have in the future.

Bibliography

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Chapter 9

Appendix

[1] <https://www.esrl.noaa.gov/psd/data/gridded/data.cpc.globaltemp.html>. CPC

Global Temperature data provided by the [NOAA/OAR/ESRL PSD](https://www.esrl.noaa.gov/psd/), Boulder, Colorado, USA, from their Web site at <https://www.esrl.noaa.gov/psd/>.